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Dysprosium doped niobium zinc fluorosilicate glasses: Interesting materials for white light emitting devices

J. Prabhakar[a], Venkata Krishnaiah Kummara[b,c,*], K. Linganna[d], P. Babu[e], C.K. Jayasankar[f], Jihoon Kim[a], V. Venkatramu[g,**]

[a] Division of Advanced Materials Engineering, Kongju National University, Cheonan, Chungnam, 31080, South Korea
[b] Laser Applications Research Group, Ton Duc Thang University, Ho Chi Minh City, Vietnam
[c] Faculty of Applied Sciences, Ton Duc Thang University, Ho Chi Minh City, Vietnam
[d] Optical Lens Research Center, Korea Photonics Technology Institute, Cheomdan venture-ro 108-9, Buk-gu, Gwangju 61007, South Korea
[e] Department of Physics, SVCR Govt. Degree College, Palamaner -517408, India
[f] Department of Physics, Sri Venkateswara University, Tirupati -517 502, India
[g] Department of Physics, Yogi Vemana University, Kadapa -516 005, India

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ABSTRACT

Trivalent dysprosium doped niobium zinc fluorosilicate glasses modified with different Nb_2O_5/ZnF_2 molar (M) ratios have been fabricated by the traditional melt-quenching technique and derived their structural, photoluminescence and decay properties using spectroscopic techniques. Induced structural modifications have been observed upon increasing content of Nb_2O_5. Maximum phonon energy of the glass matrix is found to be 1010 cm^{-1} from the Raman spectrum. The emission spectra of these glasses exhibit two intense bands at 480 and 570 nm besides a weak red emission at 650 nm. The decay profiles of Dy^{3+} ion for the $^4F_{9/2}$ level exhibit a non-exponential behavior for all the glasses. The intrinsic lifetimes for the $^4F_{9/2}$ level of Dy^{3+} ion have been determined by using the Inokuti-Hirayama model and are found to be 409, 366 and 325µs for the glasses with Nb_2O_5/ZnF_2:10/30, 20/20 and 30/10 M ratios, respectively. The color coordinates have been evaluated from the emission spectra of the glasses and found that the glass with Nb_2O_5/ZnF_2:30/10 M ratio appears near to the equal energy point. The correlated color temperature matches well to the summer sunlight region (4900–5600 K), indicating that the glasses could be a potential candidate for white light emitting devices.

1. Introduction

Trivalent dysprosium (Dy^{3+})-doped materials have potential applications in visible lasers [1,2], mid infra-red lasers [3,4], optical amplifiers [5] and white light emitting diodes (WLEDs) [6]. Of these, the WLEDs are playing a key role in solid state lighting technology and received a great attention for replacing the existing fluorescent and incandescent lamps because of their low production cost, high reliability, environmentally benign, low energy consumption and longer lifetime [7].

White light emission can be accomplished with co-doped and tri-doped systems through upconversion [8] and down-shifting [9] processes, by mixing three primary colors (Red, Green and Blue). However, it is difficult to optimize the concentrations of co-doped and tri-doped systems to achieve the optimum intensities of the red, green and blue emissions for the production of white

* Corresponding author at: Laser Applications Research Group, Ton Duc Thang University, Ho Chi Minh City, Vietnam.
** Corresponding author.
E-mail addresses: kvkrishaniah@tdtu.edu.vn (V.K. Kummara), vvramuphd@gmail.com (V. Venkatramu).

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luminescence. An alternative way is to use single Dy^{3+}-doped systems through the optimization of yellow to blue (Y/B) intensity ratio. Further, the Dy^{3+} ion has strong absorption bands in the ultra-violet (UV) region suitable to pump with low-cost GaN and InGaN diodes. Previous studies have shown that the Y/B ratio could be controlled by changing the Dy^{3+} ion concentration [10–12], pump wavelength [13,14] and glass matrix [15]. Recently, Sundara Rao et al. reported the effect of aluminum ions on Y/B ratio of Dy^{3+} ions in lead silicate glass matrix under the excitation of 352 and 451 nm [16]. Pisarska et al. studied the variation of Y/B ratio as a function of B_2O_3/PbO ratio [15].

In our earlier work, Dy^{3+} ion concentration [12] and pump wavelength [14] dependence white luminescence studies have been demonstrated and reported that the 0.1 mol% Dy_2O_3-doped glasses exhibit a better white luminescence under 387 nm excitation. However, still there is a large scope to achieve ideal white light emission from these glasses through the optimization of glass composition. Moreover, optical transmittance of the glass in the visible region was increased with the addition of Nb_2O_5 content [17]. Hence, it is interesting to study the effect of composition on white light luminescence properties. In the present study, we have fabricated Nb_2O_5 dependence zinc fluorosilicate glasses and characterized for the white light emission applications since Nb_2O_5 modified glasses exhibited a high refractive index, good transparency from visible to infrared region, high glass transition temperature, high thermo-mechanical stability, relatively high non-linear optical properties [16,18–20]. Luminescence properties of Eu^{3+}-doped niobium-borate glasses were enhanced significantly with increasing the Nb_2O_5 content [21].

Therefore, the present work focuses on the composition dependent 0.1 mol % Dy_2O_3-doped niobium zinc fluorosilicate glasses modified with different Nb_2O_5/ZnF_2 (10:30, 20:20 and 30:10) molar ratios for the better understanding and optimization of the white luminescence under the excitation of 350 and 387 nm. The Y/B intensity ratios, energy transfer parameters, chromaticity coordinates and correlated color temperatures were evaluated and compared to the glass with 20:20 M ratio of Nb_2O_5/ZnF_2 [12,14] along with other reported ones [6,22–25].

2. Material and methods

Dy^{3+}-doped niobium zinc fluorosilicate glasses with different compositions ($29.9SiO_2$–$10Nb_2O_5$–$20K_2O$–$30ZnF_2$–10LiF–$0.1Dy_2O_3$, $29.9SiO_2$–$20Nb_2O_5$–$20K_2O$–$20ZnF_2$–10LiF– $0.1Dy_2O_3$ and $29.9SiO_2$–$30Nb_2O_5$–$20K_2O$–$10ZnF_2$–10LiF–$0.1Dy_2O_3$, labeled as Nb10:ZF30, Nb20:ZF20 and Nb30:ZF10, respectively) were fabricated by the melt-quenching technique [12,14]. The precursor chemicals were thoroughly mixed in an agate mortar and then taken into a platinum crucible. The powders were then heated at 1300 °C for 3 h in a muffle furnace under ambient atmosphere. The glass melt was casted onto a brass mold preheated at 510 °C and subsequently annealed at the same temperature for 12 h in order to eliminate any residual internal stress and then allowed to cool to reach room temperature. Finally, the glass samples were polished to get optical quality for optical characterization.

Raman spectra of the glasses were measured upon 786 nm diode laser excitation by using spectrometer (Renishaw inVia Raman microscope). The emission spectra were recorded by using HORIBA JOBIN YVON Fluorolog-3 spectrofluorimeter at room temperature. Decay curves were recorded by monitoring the emission at 573 nm under the 387 nm excitation.

3. Results and discussion

Raman spectra of niobium zinc fluorosilicate glasses at different molar ratios in the region of 220–1270 cm^{-1} are shown in Fig. 1. The well resolved spectra exhibit seven Raman modes at 244, 480, 596, 681, 818, 888 and 1010 cm^{-1} for the glass with 30:10 M ratio of Nb_2O_5/ZnF_2 whereas the Raman modes at 681 and 818 cm^{-1} do not appear as resolved bands for the other two glasses. This may be due to presence of high Nb_2O_5 content. Density of the glass sample is found to be 3.44, 3.45, and 3.60 g/cm^3 for 10:30, 20:20 and

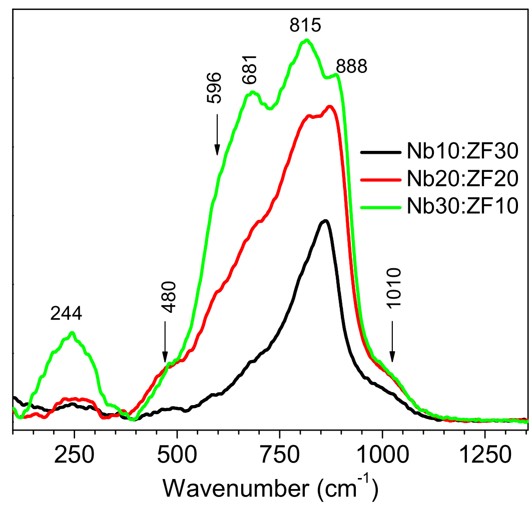

Fig. 1. Raman spectra of glasses modified with 10:30, 20:20 and 30:10 M ratios of Nb_2O_5 and ZnF_2.

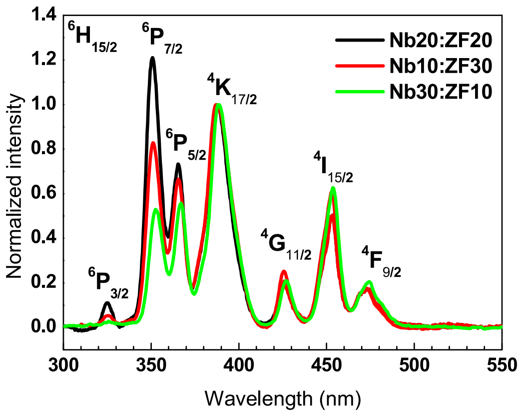

Fig. 2. Normalized (with respect to $^6H_{15/2} \rightarrow {}^6K_{17/2}$) excitation spectra of 0.1 mol % Dy^{3+}-doped niobium zinc fluorosilicate glasses for different molar ratios.

30:10 M ratio of Nb_2O_5/ZnF_2, respectively. The energy of high intense vibrational mode (phonon energy) decreases from the 888 cm^{-1} to 818 cm^{-1} with the increase in the content of Nb_2O_5 from 10 to 30 mol%, respectively. In other words, it is also perceived that the vibrational energy at 888 cm^{-1} decreases with the decrease in the content of ZnF_2 from 30 to 10, respectively. The band at 1010 cm^{-1} is not related to the Nb vibrations [26], but it could be assigned to the stretching vibrations of silicon tetrahedra. This is also supported by the decrease in the relative intensity of Raman band at 1010 cm^{-1} with increasing the molar ratio of Nb_2O_5/SiO_2 [27]. It is interesting to note that the intensity of Raman bands increases with the increase of Nb_2O_5 content for all the vibrational modes. Initially, high intense band is observed at 888 cm^{-1} and then the intensity of the band at 815 cm^{-1} is increased with increasing Nb_2O_5 content. An enhancement of 815 cm^{-1} peak intensity is an indication of increase in the Nb—O bonds and non-bridging oxygens [28]. Moreover, the other bands and a shoulder of spectra are resolved for the glass with high Nb_2O_5 content. The vibrations at 815 cm^{-1} and 888 cm^{-1} correspond to the non-bridging oxygen and Nb—O—Si, respectively, in the glass matrix [29]. This vibration chain extended to Nb—O—Si—(O—M) due to the presence of alkali ions (M═K and Li). The intermediate frequency band around at 681 cm^{-1} is attributed to the vibrational modes of Si-O-Si bonds attached to the bending vibrational modes of Nb—O bonds in NbO_6 octahedron without non-bridging oxygen [29]. The vibrational bands at 244, 480 and 596 cm^{-1} are ascribed to purely niobium octahedron (NbO_6) through bridging bonds N—O—Nb [30]. With increasing of Nb_2O_5 content in the glass matrix, there is a great enhancement of bandwidth due to the Nb-related vibrational modes [31]. Moreover, the degree of bonding is increased between various structural groups in the glass network with the increase of Nb_2O_5 content [17].

The excitation spectra were obtained by monitoring the emission at 570 nm are shown in Fig. 2. Spectra exhibit more or less similar behavior to the glass with 20:20 M ratio of Nb_2O_5/ZnF_2 [12,13] except variation in their intensities due to non-bridging oxygens as evidenced from Raman spectra. The spectra are normalized with respect to the $^6H_{15/2} \rightarrow {}^4K_{17/2}$ transition. It is noticed that the $^6H_{15/2} \rightarrow {}^6P_{7/2}$ transition possesses higher intensity for the glass with 10:30 M ratio whereas lower intensity for the glass with 30:10 M ratio.

Normalized luminescence spectra of Dy^{3+}-doped niobium zinc fluorosilicate glasses with varying Nb_2O_5/ZnF_2 ratio, obtained by exciting the glass samples at 350 nm and 387 nm and are shown in Fig. 3(a) and (b), respectively. Emission spectra were exhibited three well known bands at 486, 577 and 669 nm which are due to $^4F_{9/2} \rightarrow {}^6H_{15/2}$, $^4F_{9/2} \rightarrow {}^6H_{13/2}$ and $^4F_{9/2} \rightarrow {}^6H_{11/2}$ transitions, respectively. Emission spectra are normalized to observe the spectral variation of the electric-dipole transition (ED), $^4F_{9/2} \rightarrow {}^6H_{15/2}$ (blue). From the emission spectra, the intensities of blue and yellow emissions are found to be more or less similar for the glasses with different molar ratios. However, the blue emission is predominant for the glasses with 20:20 and 10:30 ratio of Nb_2O_5/ZnF_2 under the excitation wavelengths 350 and 387 nm, respectively. High intensity of blue emission could be due to high covalence between Dy—O bonds and the presence of Dy^{3+} ion in different sites under suitable excitation wavelength. The Y/B ratios under 350 nm excitation are found to be 1.77, 0.82 and 1.15 for the glass with 10:30, 20:20 and 30:10 M ratios of Nb_2O_5/ZnF_2, respectively. As can be seen, the blue emission is dominant for a glass with 20:20 M ratio. On the other hand, under 387 nm excitation, the Y/B ratios are found to be 0.56, 0.86, and 0.92 for the glasses with 10:30, 20:20 and 30:10 M ratios of Nb_2O_5/ZnF_2, respectively, and higher blue emission is observed for 10:30 M ratio. The higher magnitudes of Y/B ratio indicate higher degree of covalency between Dy^{3+} ions and their ligand environment. The Y/B ratios (1.15 and 0.92) are close to unity for a glass with 30:10 M ratios of Nb_2O_5/ZnF_2 under both 350 and 387 nm excitations. This could be due to the presence of higher non-bridging oxygens and/or lower phonon energy around the Dy^{3+} ions. Higher value of Y/B ratio indicates that the higher concentrations of Dy^{3+} ions may be dissociated from Dy—O—Dy bonds and the possibility for high degree of de-polymerization in the glass matrix [32]. In our glass system, high value of Y/B ratio of 1.77 for the glasses with 10:30 ratio of Nb_2O_5/ZnF_2 indicates a large amount of dissociated Dy^{3+} ions from Dy—O—Dy bonds. This could be due to the increase of non-bridging oxygens or the formation of the Dy—F—Dy or Nb—F—Nb bonds (decrease of Dy—O—Dy or Nb—O—Nb bonds) in the glass matrix. This is also evidenced from the Raman mode at 888 cm^{-1} which purely related to the Nb—F(O) vibrations. The Y/B ratio of 0.1 mol% Dy^{3+}-doped glasses are compared to those of other Dy^{3+}:systems in Table 1 and are found to be comparable to LBO:Dy glass (0.93) [6] and lower than silicate glass (1.96) [24].

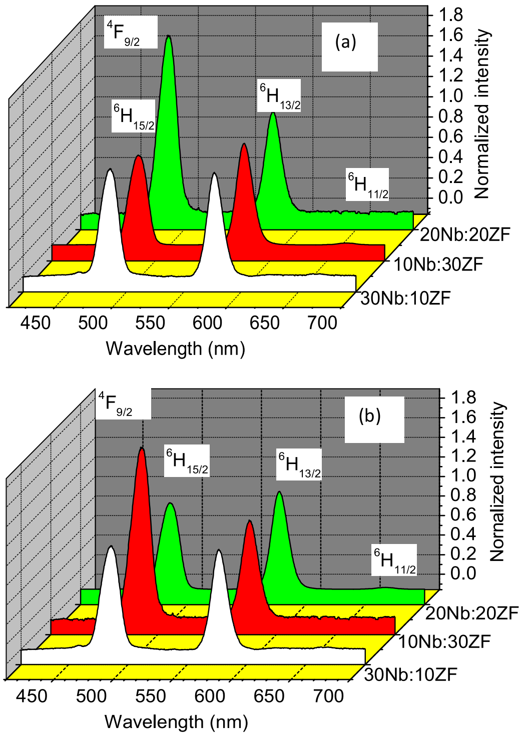

Fig. 3. Luminescence spectra of 0.1 mol % Dy^{3+} -doped niobium zinc fluorosilcate glasses under (a) 350 nm and (b) 387 nm excitations. The spectra normalized to the maximum intensity of the $^4F_{9/2}$→$^6H_{13/2}$ transition and compared with Nb_2O_5/ZnF_2: 20/20 M ratio.

Table 1
Comparison of Y/B ratios, (*x*,*y*) chromaticity color coordinates, correlated color temperature (CCT, K) of Dy^{3+}:systems under 350 and 387 nm excitations.

Glass system (Dy^{3+} ion concentration)	Y/B ratio		(*x*,*y*)		CCT		Reference
	350 nm	387 nm	350 nm	387 nm	350 nm	387 nm	
Nb20:ZF20 (0.1 mol %)	0.82	0.86	(0.331, 0.374)	(0.332, 0.371)	5556	5520	Present work
Nb10:ZF30 (0.1 mol %)	1.77	0.56	(0.352, 0.388)	(0.282, 0.329)	4891	8267	
Nb30:ZF10 (0.1 mol %)	1.15	0.90	(0.333, 0.377)	(0.332, 0.372)	5485	5520	
LBO:Dy (0.25 mol%)	0.93	0.95	(0.342, 0.371)	(0.344, 0.371)	5169*	5101*	[6]
Silicate (0.5 mol %)	1.96	–	(0.381,0.434)	–	4324*	–	[23]
Oxyfluoride (0.25 mol%)	–	–	(0.23,0.31)	(0.38,0.40)	13,750	4163	[24]
SLBPDy (1.0 mol%)	–	0.72	–	(0.45,0.42)	–	2115	[36]
Sodium Borate (1.5 mol%)	–	–	–	(0.44, 0.50) @ 375nm excitation	–	3756 @ 375nm excitation	[37]
TSWD2 (0.2 mol%)	–	1.51@ 452 nm	–	(0.35,0.40) @ 452 nm	–	4971 @ 452 nm	[38]

* Indicates the values are evaluated from the data mentioned in the papers.

The luminescence decay profiles of Dy^{3+} ions for the $^4F_{9/2}$ level under 387 nm excitation are shown in Fig. 4 and are found to be non-exponential. The lifetime for the $^4F_{9/2}$ level of Dy^{3+} ion is evaluated from the equation, $\langle\tau\rangle = \frac{\int tI(t)dt}{\int I(t)dt}$ and is found to be 281, 265 and 245 µs for the glasses with 20:20, 10:30 and 30:10 M ratios of Nb_2O_5/ZnF_2, respectively. It is noticed that the lifetime decreases with varying Nb_2O_5/ZnF_2 molar ratio due to the increase in non-radiative processes of Dy^{3+} ions. The lower lifetime for 30:10 of Nb_2O_5/ZnF_2 ratio indicates more covalent Dy—O bonds in the glass network as evidenced from the Raman spectra, leading to increase in the intensity and broadening of vibrational frequencies. This can be understood from the Raman analysis. The non-exponential decay profiles are analyzed under the frame of Inokuti-Hirayama (IH) model [33] by taking the intrinsic lifetime, τ_0 as 0.41 ms [12] found for 20:20 M ratio. According to this model, the luminescence intensity *I*(t) of the de-excited ion is given by

$$I(t) = I_0 \exp\left[-\left(\frac{t}{\tau_0}\right) - Q\left(\frac{t}{\tau_0}\right)^{3/S}\right] \tag{1}$$

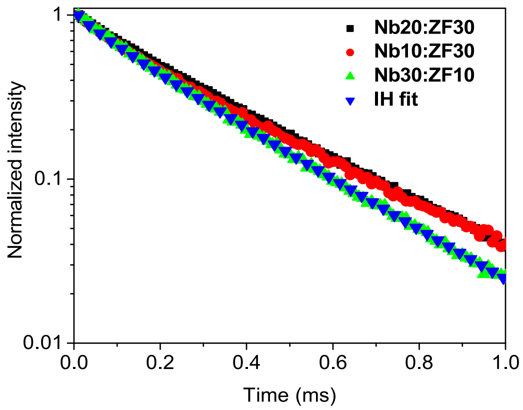

Fig. 4. Decay curves for the $^4F_{9/2}$ level of Dy^{3+} ion under 387 nm excitation along with IH fit for S = 6.

where τ_0 is the inherent decay time of the donors without the acceptors, S (=6, 8 or 10) relates the mechanism of the interaction, i.e. dipole–dipole, dipole-quadrupole or quadrupole-quadrupole, respectively and Q is the energy transfer parameter. The decay curves are fitted to the Eq. (1) by varying τ_0 and Q. The best fit is obtained for $S = 6$, indicating that the nature of interaction between Dy^{3+} ions is dipole-dipole type. From the fitting, Q values are estimated to be 0.54, 0.27 and 0.34 for the glasses with 10:30, 20:20, and 30:10 M ratios of Nb_2O_5/ZnF_2, respectively. It is noticed that the lifetime of Dy^{3+} for the $^4F_{9/2}$ level decreases with increasing Nb_2O_5 content, due to increase in non-radiative processes.

The combination of blue and yellow emissions from the Dy^{3+}-doped material can produce a white luminescence. Dy^{3+}-doped glasses can be considered as an attractive materials for two-color phosphors since Dy^{3+} ion exhibits intense emissions at blue (482 nm, $^4F_{9/2} \rightarrow {}^6H_{15/2}$) and yellow (573 nm, $^4F_{9/2} \rightarrow {}^6H_{13/2}$) regions. It is well known that the $^4F_{9/2} \rightarrow {}^6H_{13/2}$ transition of Dy^{3+} ions is hypersensitive and its intensity strongly depends on the nature of the host, whereas the intensity of magnetic-dipole transition, $^4F_{9/2} \rightarrow {}^6H_{15/2}$ is less sensitive to the host. Therefore, the intensity ratio (Y/B) of these transitions can be tailored such that the Dy^{3+}-doped materials would generate white light. The white light produced by Dy^{3+}-doped glass samples can be analyzed in the frame of the chromaticity coordinates of colors. The procedure established by the Commission International de I'Eclairage (CIE) is the most widely used method to describe the composition of any color in terms of three primary colors i.e. Red, Green and Blue (RGB). Artificial "colors", denoted by X, Y, Z, also called tristimulus values, can be added to produce real spectral colors (RGB). The chromaticity color coordinates are obtained from the following equations:

$$x = \frac{X}{X+Y+Z};\ y = \frac{Y}{X+Y+Z};\ \text{and}\ z = \frac{Z}{X+Y+Z} \tag{2}$$

where X, Y, and Z are the tristimulus values given by: $X = \int E(\lambda)\bar{x}(\lambda)d\lambda$ $Y = \int E(\lambda)\bar{y}(\lambda)d\lambda$ and $Z = \int E(\lambda)\bar{z}(\lambda)d\lambda$; $\bar{x}$, $\bar{y}$, and $\bar{z}$ are matching functions of CIE, to define the chromatic characteristics of a visible source and they can be thought of as the spectral sensitivity curves of three linear light detectors yielding the CIE tristimulus values X, Y and Z.

The CIE chromaticity color coordinates are evaluated from the emission spectra and are presented in Figs. 5(a) and (b) under 387 and 350 nm excitations, respectively. The McCamy equation has been used to compute correlated color temperature (CCT) [34,35] from the CIE chromaticity coordinates. The (x, y) chromaticity coordinates and CCT values are found to be (0.282, 0.329), (0.332, 0.372) and (0.332, 0.371) and, 8267, 5520 and 5520 K for the glasses with 10:30, 20:20 and 30:10 M ratios of Nb_2O_5/ZnF_2, respectively, under 387 nm excitation. On the other hand, under 350 nm excitation, the (x, y) chromaticity coordinates and CCT are found to be (0.352, 0.388), (0.331, 0.374) and (0.333, 0.377) and 4891, 5556 and 5485 K for the glasses with 10:30, 20:20 and 30:10 M ratios, respectively. The CCT values, obtained under 350 nm excitation, are comparable to those of lithium borate (0.25 mol %, 5169 K) [6], silicate (0.5 mol%, 4324 K) [24] glasses (which are evaluated from the reported CIE chromaticity coordinates), apart from the reported oxyfluoride (0.25 mol%, 13,750 K) [24] glass, which falls under clear blue sky light source. Under 387 nm excitation, the CCT values for the investigated glasses are higher than those of other reported glasses including LBO:Dy (5101 K) [6], oxyfluoride (2115 K) [22], SZBPDy (3756 K) [36], sodium borate (3756 K) [37] and TSWD2 (4971 K) [38]. The CCT values of the investigated glasses match well in the region of the light sources such as summer sunlight (4900–5600 K) and clear blue sky (8000–27000 K). The above results indicate that these glasses find potential applications for the development of summer sunlight white light sources.

4. Conclusions

Luminescence properties of Dy^{3+}-doped oxyfluoride glasses modified with different Nb_2O_5/ZnF_2 molar ratios have been studied under 350 and 387 nm excitations for the white luminescence. The maximum phonon energy is found to be 1010 cm^{-1} for a glass with high content of Nb_2O_5 content. The emission spectra exhibited two intense bands at 480 and 570 nm corresponding to the $^4F_{9/2} \rightarrow {}^6H_{15/2}$ (blue) and $^4F_{9/2} \rightarrow {}^6H_{13/2}$ (yellow) transitions, respectively, besides a weak red emission attributed to the $^4F_{9/2} \rightarrow {}^6H_{11/2}$ transition. Higher Y/B ratio have been noticed for the glass with 10:30 M ratio of Nb_2O_5/ZnF_2 compared to the other glasses of the

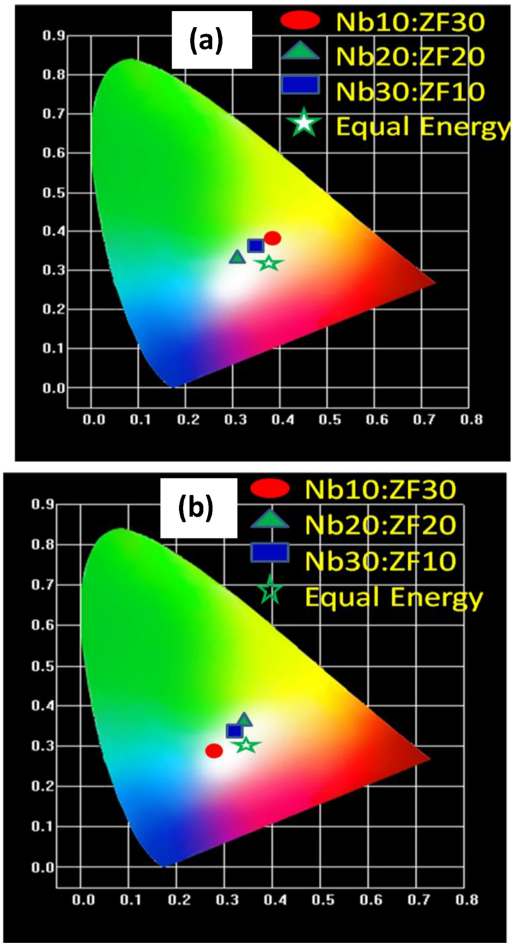

Fig. 5. Representation of CIE chromaticity color coordinates of Dy^{3+}-doped Nb_2O_5/ZnF_2:20/20 M ratio glasses under (a) 350 nm and (b) 387 nm excitations.

present study. The decay curve analysis reveals that the mechanism of interaction for the energy transfer between the Dy^{3+} ions is of dipole-dipole type. The color coordinates and correlated color temperature for the glass with 30:10 (Nb_2O_5/ZnF_2) molar ratio is found to be nearer to the ideal white light and hence, it could be a potential candidate for the generation of white light sources under 350 and 387 nm excitations.

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